

Monitoring System for Aquaponics: Case Study for Herbs Cultivation

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Abstract: A monitoring system is implemented in an aquaponics setup to enhance the growth of medicinal herbs such as *Chlorophytum comosum* (Spider Plant), Rosemary, Oregano, and other beneficial medicinal plants. *Chlorophytum Comosum* (Spider Plant) : known for air-purifying properties and traditional medicinal uses and it can be used to treat respiratory issues and improve skin health. Rosemary (*Salvia Rosmarinus*) : rich in antioxidants, improves memory, and supports digestion and it can be used in traditional medicine for anti-inflammatory and antibacterial benefits. Oregano (*Origanum Vulgare*) : contains antibacterial, antiviral, and antifungal properties and traditionally used to support respiratory and immune health. This system integrates IoT sensors, remote monitoring technologies, and AI-driven analytics to precisely track water quality, nutrient levels, pH, dissolved oxygen, and temperature, ensuring optimal conditions for both plants and aquatic life. A cloud-based dashboard allows real-time monitoring, enabling immediate adjustments and improving efficiency. This dashboard collects data from multiple IoT sensors and cameras, displaying key parameters such as water quality, temperature, pH levels, and plant health status in a user-friendly interface. Users can access the dashboard from any internet-connected device, and adjust system settings remotely. High-resolution cameras capture medicinal plant images for health analysis. Key Benefits are (1) Optimized water usage and real-time automated adjustments, reducing wastage. (2) Prevention of crop loss and production of high-quality, pesticide-free herbs. (3) Remote monitoring and automation improve system efficiency and reduce labour. (4) Enhanced plant health through AI-driven monitoring and automated nutrient balancing. (5) Predictive maintenance reduces downtime, improves operational reliability and system efficiency. (6) Supports sustainable, organic urban farming and commercial-scale agriculture.